

LM04 Analyzing Statements of Cash Flows I

1. Introduction.....	2
2. Linkages Between the Financial Statements.....	2
3. The Direct Method for Cash Flows From Operating Activities	3
4. The Indirect Method for Cash Flows from Operating Activities.....	5
5. Conversion from the Indirect to Direct Method	6
6. Cash Flows from Investing Activities.....	6
7. Cash Flows from Financing Activities.....	7
8. Differences in Cash Flow Statements Prepared Under US GAAP Versus IFRS.....	8
Summary.....	10

Required disclaimer: IFT is a CFA Institute Prep Provider. Only CFA Institute Prep Providers are permitted to make use of CFA Institute copyrighted materials which are the building blocks of the exam. We are also required to create / use updated materials every year and this is validated by CFA Institute. Our products and services substantially cover the relevant curriculum and exam and this is validated by CFA Institute. In our advertising, any statement about the numbers of questions in our products and services relates to unique, original, proprietary questions. CFA Institute Prep Providers are forbidden from including CFA Institute official mock exam questions or any questions other than the end of reading questions within their products and services.

CFA Institute does not endorse, promote, review or warrant the accuracy or quality of the product and services offered by IFT. CFA Institute®, CFA® and “Chartered Financial Analyst®” are trademarks owned by CFA Institute.

© Copyright CFA Institute

Version 1.0

1. Introduction

The cash flow statement provides important information about a company's cash receipts and payments during an accounting period. It is a vital information source that assists users to evaluate a company's liquidity, solvency, and financial flexibility.

Under both IFRS and US GAAP, cash flows are categorized as operating, investing, or financing activities on the cash flow statement.

Operating activities: These are activities related to the normal operations of a company. Examples include:

- Cash inflows such as cash collected from sales, commissions, royalties, etc.
- Cash outflows such as cash payments for inventory, salaries, and operating expenses.
- Cash payments and receipts related to trading securities (securities that are not bought as investments).

Investing activities: These are activities associated with acquisition and disposal of long-term assets. Examples include:

- Cash from sale of property, plant, and equipment.
- Cash spent to purchase property, plant, and equipment.
- Cash payments and receipts related to investment securities (not trading securities).

Financing activities: These are activities related to obtaining or repaying capital. Examples include:

- Issuance or repurchase of a company's own preferred or common stock.
- Issuance or repayment of debt.
- Dividend payments to shareholders.

2. Linkages Between the Financial Statements

Link between the Cash Flow Statement and the Balance Sheet

Cash is an asset and is reported on the balance sheet. The cash flow statement explains the change in cash during an accounting period. This can be illustrated through a simple scenario. Assume that beginning cash is 1,100. This is reported on the balance sheet. The cash flow statement will show the cash receipts and cash payments. For the scenario presented below, the cash receipts equal 3,200 and the cash payments equal 3,300 which means that the net change in cash is -100. This explains how the cash balance went from 1,100 at the start of the period to 1,000 at the end of the period.

<i>Beginning Balance Sheet 1 Jan 2015</i>	<i>Statement of Cash Flows for Year Ended 31 December 2015</i>		<i>Ending Balance Sheet at 31 Dec 2015</i>
Beginning Cash	Plus: Cash Receipts	Less: Cash Payments	Ending Cash

1,100	3,200	3,300	1,000
-------	-------	-------	-------

Link between Cash Flow Statement, Balance Sheet, and Income Statement

Let's consider an example of how items on the balance sheet are related to the income statement and the cash flow statement. Suppose the beginning accounts receivable is 200, the revenue during the year is 5,000 and the cash collected from customers is 4,800. What is the ending accounts receivables? The table below makes it easy to compute the missing amount. We see that the ending accounts receivables will be 400.

<i>Balance Sheet at 1 Jan 2015</i>	<i>Income Statement</i>	<i>Statement of Cash Flows</i>	<i>Balance Sheet at 31 Dec 2015</i>
Beginning A/R	Plus: Revenue	Less: Cash Collected from Customers	Ending A/R
200	5,000	4,800	400

This example clearly shows that receivables (balance sheet item), revenue (income statement item) and cash collected from customers (cash flow item) are related as follows:

$$\text{Ending receivables} = \text{Beginning receivables} + \text{Revenue} - \text{Cash collected from customers}$$

3. The Direct Method for Cash Flows From Operating Activities

Under IFRS and US GAAP, there are two acceptable formats for reporting cash flow from operating activities: indirect and direct.

- The **indirect method** shows how cash flow from operations can be obtained from reported net income through a series of adjustments.
- The **direct method** shows the specific cash inflows and outflows that result in reported cash flow from operating activities.

Indirect Format Sample

With the indirect method, we start with net income and make several adjustments for non-cash, non-operating items to arrive at the cash flow from operations. Shown below is a sample of the indirect format for a fictitious company called K2 Corp.

Net income	2,775
Depreciation	1,000
Gain on sale of equipment	(200)
Increase in accounts receivable	(150)
Increase in inventory	(600)
Increase in pre-paid expenses	(30)
Increase in accounts payable	300

Increase in wages payable	10
Increase in tax payable	5
Increase in other accrued liabilities	100
Decrease in interest payable	(10)
<i>Cash flow from operations</i>	<i>3,200</i>

Direct Format Sample

In the direct format, we look at the specific cash inflows and outflows that resulted in cash flow from operating activities. This method is encouraged by both IFRS and US GAAP.

Cash from customers	24,850
Cash paid to suppliers	(10,300)
Cash paid to employees	(7,990)
Cash paid for other operating expenses	(1,930)
Cash paid for interest	(510)
Cash paid for taxes	(920)
<i>Cash flow from operations</i>	<i>3,200</i>

Notice that while the presentation formats are different, the cash flow from operations number is the same under both methods.

Operating Activities: Direct Method

Only cash flow from operating activities is presented differently under the two methods. Presentation of cash flow from investing activities and cash flow from financing activities is the same under both methods.

Direct method:

In the direct method, we take each item from the income statement and convert it to its cash equivalent by removing the impact of accrual accounting.

The rules to adjust are:

- Increase in assets is use of cash (-ve adjustment) and decrease in asset is source of cash (+ve adjustment).
- Increase in liability is source of cash (+ve adjustment) and decrease in liability is use of cash (-ve adjustment).

Cash collected from customers: Adjust sales for changes in accounts receivable and unearned revenue.

Cash for inputs: Adjust COGS for changes in inventory and accounts payable.

Cash operating expenses: Adjust SG&A for changes in related accrued liabilities or prepaid

expenses.

Cash interest paid: Adjust interest expense for changes in interest payable.

Cash taxes paid: Adjust tax expense for changes in tax payable and changes in deferred tax assets and liabilities.

Example

Consider a company that reported sales of \$10 million. Accounts receivable for the year went up from \$2 million to \$4 million. Unearned revenue went up from \$1 million to \$2 million. Calculate cash collected from customers.

Solution:

Δ Accounts receivable = \$2 million. This is an asset and increase in asset is use of cash so -ve adjustment.

Δ Unearned revenue = \$1million. This is a liability and increase in liability is source of cash so +ve adjustment.

Cash collected from customers = + \$10 million - \$2million + \$1million = \$9 million.

Example

Consider a company with COGS of \$20 million for a particular period. During this period inventory increased by \$4 million and accounts payable went up by \$2 million. Calculate the cash paid for inputs.

Solution:

Δ Inventory = \$4 million. This is an asset and increase in asset is use of cash so -ve adjustment.

Δ Accounts payable = \$2 million. This is a liability and increase in liability is source of cash so +ve adjustment.

Cash paid for inputs = - \$20 million - \$4 million + \$2 million = - \$22 million

4. The Indirect Method for Cash Flows from Operating Activities

Indirect method shows how cash flow from operations can be obtained from reported net income as a result of a series of adjustments.

The steps are:

- Begin with net income.
- Add back all non-cash charges to income and subtract all non-cash components of revenue (For example, add depreciation and amortization).
- Subtract any gains that resulted from financing or investing cash flows (For example,

gain on the sale of an equipment).

- Add or subtract changes to related balance sheet operating accounts.
- Decrease in operating assets (source of cash) should be added and increase in operating assets (use of cash) should be subtracted.
- Similarly, increase in current liabilities (source of cash) should be added and decrease in current liabilities (use of cash) should be subtracted.

Example

Consider a company with net income of \$100 million in 2001. Depreciation expense is \$10 million. Gain on sale of equipment is \$4 million. Increase in A/R is \$8 million. Increase in A/P is \$4 million. Increase in inventory is \$10 million. Calculate CFO using the indirect method.

Solution:

Net income	+100
Add non-cash charges (depreciation)	+ 10
Less gain on sale of equipment	- 4
Less increase in A/R	-8
Add increase in A/P	+4
Less increase in inventory	<u>-10</u>
Total	92
CFO = \$ 92 million	

5. Conversion from the Indirect to Direct Method

Instructor's note: The probability of getting tested on this topic on the exam is low.

The operating cash flow from indirect method can be converted to direct by using the three-step process:

- Aggregate all the revenues and expenses.
- Remove all non-cash items from aggregated revenues and expenses and break up remaining items into relevant cash flow items.
- Convert accrual amounts to cash flow amounts by adjusting for changes in corresponding working accounts.

6. Cash Flows from Investing Activities

CFI is calculated by examining the change in the gross asset account that results from investing activities. Typically, this change results from purchases or sale of equipment (long-term assets). To determine the cash inflow from the sale of equipment, we need to use the expression shown below.

$$\text{Cash from sale of equipment} = \text{Historical cost of equipment sold}$$

– Accumulated depreciation on equipment sold
+ Gain on sale of equipment

where:

Historical cost of equipment sold = beginning balance

+ equipment purchased

– ending balance equipment

Accumulated depreciation on equipment sold = beginning value of depreciation + depreciation expense – ending value of depreciation

Example

The balance sheet extract for Jackal Labs Ltd shows the machinery and accumulated depreciation balances for the years 2011 and 2012.

	2011	2012
Machinery (Gross)	\$80 million	\$91 million
Accumulated depreciation	\$25 million	\$31 million

Further information provided is as follows:

Gain on sale of machinery \$1.5 million

Depreciation expense for 2012 \$7 million

Capital expenditure on machinery \$14 million

What is the cash received from sale of equipment?

Solution:

Cash from sale of machinery = historical cost of equipment sold – accumulated depreciation on equipment sold + gain on sale of equipment

We know the gain is \$1.5 million. Calculate the other components in the equation:

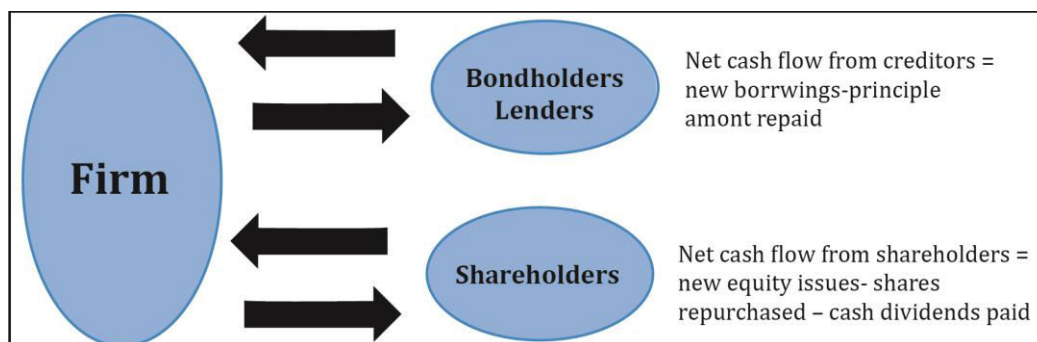
Historical cost of equipment sold = beginning balance + equipment purchased – ending balance of equipment = 80 + 14 – 91 = \$3 million

Accumulated depreciation on equipment sold = beginning value of depreciation + depreciation expense – ending value of depreciation = 25 + 7 – 31 = \$1 million

Cash from sale of machinery = 3 – 1 + 1.5 = 3.5

7. Cash Flows from Financing Activities

Cash flow from financing activities refers to cash flows between the firm and the suppliers of capital. Suppliers of capital include creditors, bondholders, and shareholders. Similar to investing activities, the presentation of cash flows from financing activities is also identical under both methods. The figure below summarizes the calculation of net cash flows from creditors, bondholders, and shareholders.



It can be calculated using the following formulae:

(1) $CFF = \text{Net cash flow from creditors} + \text{Net cash flow from shareholders}$

(2) $\text{Net cash flow from creditors} = \text{New borrowings} - \text{Principal repaid}$

(3) $\text{Net cash flow from shareholders} = \text{New equity issued} - \text{Shares repurchased} - \text{Cash dividends}$

Example

The following information is available about company ABC for 2001.

New borrowings	\$10 million
Principal repaid	\$5 million
New equity issued	\$5 million
Shares repurchased	-
Dividends paid	\$2 million

Calculate CFF.

Solution:

$\text{Net cash flow from creditors} = \text{New borrowings} - \text{Principal repaid} = 10 - 5 = \5 million

$\text{Net cash flow from shareholders} = \text{New equity issued} - \text{Shares repurchased} - \text{Cash dividends} = 5 - 0 - 2 = \3 million

$CFF = \text{Net cash flow from creditors} + \text{Net cash flow from shareholders} = 5 + 3 = \8 million

8. Differences in Cash Flow Statements Prepared Under US GAAP Versus IFRS

The reporting of interest paid/received and dividends paid/received is different between IFRS and US GAAP. The differences between the two standards are summarized in the table below.

Cash flow	IFRS	US GAAP
-----------	------	---------

Interest received	Operating or investing	Operating
Interest paid	Operating or financing	Operating
Dividends received	Operating or investing	Operating
Dividends paid	Operating or financing	Financing

In addition to the points made above, IFRS and US GAAP also have some differences with respect to bank overdrafts, taxes paid, and the format of the cash flow statement. These are outlined in the table below.

Cash Flow	IFRS	US GAAP
Bank overdrafts	Considered part of cash equivalents.	Not considered part of cash equivalents and classified as financing.
Taxes paid	Generally categorized as operating, but a portion can be allocated to investing or financing if it can be specifically identified with these categories.	Operating.
Format of statement	Both direct and indirect formats are allowed but the direct format is encouraged.	Both direct and indirect formats are allowed but the direct format is encouraged. A reconciliation of net income to cash flow from operating activities must be provided regardless of method used.

Summary

LO: Describe how the cash flow statement is linked to the income statement and the balance sheet.

Cash is an asset. The cash flow statement ultimately shows the change in cash during an accounting period. The beginning and ending balances of cash are shown on the balance sheet and the bottom of the cash flow statement reconciles beginning cash with ending cash.

Because a company's operating activities are reported on an accrual basis in the income statement, any differences between the accrual basis and cash basis for accounting result in an increase or decrease in some asset or liability on the balance sheet.

LO: Describe the steps in the preparation of direct and indirect cash flow statements, including how cash flows can be computed using income statement and balance sheet data.

CFO can be computed using the direct method or the indirect method.

Direct method:

In the direct method, we take each item from the income statement and convert it to its cash equivalent by removing the impact of accrual accounting.

The rules to adjust are:

- Increase in assets is use of cash (-ve adjustment) and decrease in asset is source of cash (+ve adjustment).
- Increase in liability is source of cash (+ve adjustment) and decrease in liability is use of cash (-ve adjustment).

Indirect method:

Indirect method shows how cash flow from operations can be obtained from reported net income as a result of a series of adjustments.

The steps are:

- Begin with net income.
- Add back all non-cash charges to income and subtract all non-cash components of revenue (For example, add depreciation and amortization).
- Subtract any gains that resulted from financing or investing cash flows (For example, gain on the sale of an equipment).
- Add or subtract changes to related balance sheet operating accounts.
- Decrease in operating assets (source of cash) should be added and increase in operating assets (use of cash) should be subtracted.
- Similarly, increase in current liabilities (source of cash) should be added and decrease in current liabilities (use of cash) should be subtracted.

CFI is calculated by determining changes in the gross asset account that result from the purchase or sale of equipment.

CFF is the sum of net cash flows from creditors and net cash flows from shareholders.

LO: Demonstrate the conversion of cash flows from the indirect to direct method.

The operating cash flow from indirect method can be converted to direct by using the three-step process:

- Aggregate all the revenues and expenses.
- Remove all non-cash items from aggregated revenues and expenses and break up remaining items into relevant cash flow items.
- Convert accrual amounts to cash flow amounts by adjusting for working capital changes.

LO: Contrast cash flow statements prepared under International Financial Reporting Standards (IFRS) and US generally accepted accounting principles (US GAAP).

Cash flow	IFRS	U.S. GAAP
Interest received	Operating or investing	Operating
Interest paid	Operating or financing	Operating
Dividends received	Operating or investing	Operating
Dividends paid	Operating or financing	Financing